Problem 4. (**Randomized SVD Algorithms**, 25 points)

The Singular Value Decomposition (SVD) is a fundamental tool in dimensionality reduction, data compression, and low-rank modeling. However, its computational cost $O(mn^2)$ for an $m \times n$ matrix with $m \geq n$——makes it impractical for modern large-scale datasets (e.g., high-resolution images or user-item rating matrices). Randomized SVD (rSVD) addresses this limitation by using random projections to approximate the dominant subspace of a matrix. It achieves near-optimal accuracy at a fraction of the time and memory cost, making it ideal for "large but approximately low-rank" data. In this experiment, you will implement and evaluate rSVD on synthetic and real-world data to understand its efficiency, accuracy, and practical utility. **Implement rSVD in MATLAB and answer the following two questions.**

Assume input matrix $X \in \mathbb{R}^{m\times n}$. The basic goal is to find a low-rank approximation of X, i.e. $||X - QQ^TX||_2 \leq \epsilon$. Two step to execute the randomized SVD. Define two mappings, $\mathcal{A}(X) : \mathbb{R}^{m\times n} \rightarrow \mathbb{R}^{m\times k}$ that approximates the range of input matrix X, $\mathcal{S}(Q) : \mathbb{R}^{m\times k} \rightarrow \mathbb{R}^{m\times m}$ that approximates the SVD of Q (k is always selected by summing the target rank and oversampling parameter p).

- $Q = \mathcal{A}(X)$. Firstly, select a Gaussian test matrix $\Omega \in \mathbb{R}^{n\times k}$. Secondly, compute $Y = A\Omega \in \mathbb{R}^{m\times k}$ ($k << m$). Thirdly, perform q times QR decomposition to Y. Forthly, orthogonalize the columns of Y to form $Q \in \mathbb{R}^{m\times k}$.
- $U = \mathcal{S}(Q)$. Form a smaller matrix $B = Q^TA \in \mathbb{R}^{k\times n}$, the SVD of matrix B is $\hat{U}\Sigma V^T$, and compute the singular vectors $U = Q\hat{U} \in \mathbb{R}^{m\times k}$.

1) **Efficiency Advantage.** Compare the runtime of rSVD and classical SVD on dense matrices of increasing size:
   - Matrix dimensions: 1000×1000, 3000×3000, and 5000×5000.
   - Fixed parameters: target rank $k = 50$, oversampling $p = 10$, power iterations $q = 2$.
   - Measure wall-clock time using `tic/toc`.
   - Plot "Matrix Size vs. Runtime" and compute the speedup ratio (i.e. the ratio of the runtime of classical SVD to that of rSVD) of rSVD over classical SVD.

   (10 points)
2) **Accuracy and Parameter Sensitivity.** Use a high-resolution grayscale image (e.g., We use 512×512 test images such as `cameraman.tif` (MATLAB built-in).) to study how reconstruction quality depends on rSVD parameters:
   - Fix target rank $k = 30$.
   - Vary oversampling $p \in \{0, 5, 10, 20\}$.
   - Vary power iterations $q \in \{0, 1, 2, 3\}$.
   - For each setting, report:
     - Relative Frobenius error: $\|A - A_k\|_F / \|A\|_F$
     - PSNR (Peak Signal-to-Noise Ratio), defined as

$$\text{PSNR} = 10 \log_{10} \left(\frac{L^2}{\text{MSE}} \right) \quad \text{(in dB)},$$

where L is the maximum possible pixel value (e.g., $L = 255$ for 8-bit images) and $\mathrm{MSE} = \frac{1}{mn}\|A - A_k\|_F^2$ is the mean squared error between the original image A and its reconstruction A_k of size $m \times n$.

- Visualize the best and worst reconstructed images.

(15 points)

Solution: